

Practice Task

Assignment Week 4B – Retrieving data from TCGA via the GDC Portal

Based on this week's lessons on **data retrieval strategies**, complete the following task and submit your answers by pasting them in the space provided.

Dataset: TCGA-COAD (Colon Adenocarcinoma) CNV & RNA-Seq

Instructions: Go to:

https://portal.gdc.cancer.gov/analysis_page?app=CohortBuilder&tab=general

STEPS:

1. Under Project, select: TCGA-COAD

2. Select the following under 'Available Data':

- a. Data Category: Copy Number Variation
- b. Data Type: Copy Number Segment
- c. Experimental Strategy: WGS

➤ **Question:**

- How many cases?

3. Go to 'Repository'

4. Repeat steps 2a, b and c.

- **Additional* → *Under Access* (left side pane): Select 'Open'

➤ **Question:**

- What's the total number of files you arrived at?

5. Select and add the following files to cart (at least 2).

- *TCGA-AA-3858-01A-01D-A91V-36.WholeGenome.RP-1657.cr.igv.reheader.seg.txt*
- *TCGA-A6-2682-01A-01D-A91U-36.WholeGenome.RP-1657.cr.igv.reheader.seg.txt*

➤ **Question:**

- What's the file format of both files?

6 Download zipped file by clicking 'Cart' (instead of 'manifest').

7. Unzip in your Windows/ OS file manager.

8. Take screenshot of your unzipped file folder.

ANSWERS

2. 265 Cases

The screenshot shows the GDC Analysis Center Cohort Builder interface. At the top, there are navigation tabs for Analysis Center, Projects, Cohort Builder, and Repository. The Cohort Builder tab is active, showing a cohort named 'Unsaved_Cohort' with 265 cases. The cohort is filtered by PROJECT (TCGA-COAD), DATA CATEGORY (copy number variation), DATA TYPE (Copy Number Segmentation), and EXPERIMENTAL STRATEGY (WGS). The interface includes a sidebar with filters for General, Demographic, General Diagnosis, Disease Status and History, Disease Specific Classifications, and Treatment. The main area displays a table of cohort filters with columns for Name, Cases, and Percentage. The filters are: Data Category (copy number variation, 265 (0.53%)), Data Type (Allele-specific Copy Number Segment, 23 (0.05%); Copy Number Segment, 265 (0.53%); Gene Level Copy Number, 23 (0.05%); Intermediate Analysis Archive, 265 (0.53%)), Experimental Strategy (Genotyping Array, 460 (0.92%); WGS, 265 (0.53%)), Workflow Type (GATK4 CNV, 265 (0.53%)), Data Format (txt, 265 (0.53%)), and Platform (Illumina, 265 (0.53%)).

4.7099 files

The screenshot shows the GDC Analysis Center Repository interface. At the top, there are navigation tabs for Analysis Center, Projects, Cohort Builder, and Repository. The Repository tab is active, showing a list of files. The interface includes a sidebar with filters for Workflow Type (ABSOLUTE LiftOver, 255 (0.02%)). The main area displays a table of files with columns for Access, File Name, Cases, Project, and Data Category. The files are: nationwidechildrens.org_biospecimen.TCGA-AZ-4313.xml (1 case, TCGA-COAD, Biospecimen), TCGA-COAD_45a09de8-b564-4737-b360-09bbaa023ab5.ascats3.gene_level_copy_number.v36.tsv (1 case, TCGA-COAD, Copy Number Variation), and DADOS_p_TCGAb3_85_86_87_88_NSP_GenomeWideSNP_6_A02_1464720.nocnv_qrch38.seq.v2.txt (1 case, TCGA-COAD, Copy Number Variation). The total number of files is 7,099, with 265 cases and 261.51 GB of data.

5. File format of both files- .txt format.

8. Screenshot of downloaded zip files

